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Journal of the Physical Society of Japan
Vol. 65, No. 3, March, 1996, pp. 695-698

LETTERS

Cu Nuclear Spin-Lattice Relaxation Study of A- and B-Sites in $La_{1.80}Sr_{0.20}CuO_4$ (T_c= 30 K)

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(Received October 25, 1995)

We report the temperature(T)-dependence of the planar ^{63}Cu nuclear spin-lattice relaxation rate, $1/T_1$, both at the Cu(A)-site (77.3 K $\leq T \leq$ 500 K) and at the Cu(B)-site (77.3 K $\leq T \leq$ 300 K) in the high-T_c Cu oxide, $La_{1.80}Sr_{0.20}CuO_4$ ($T_c \sim$ 30 K), measured by the Cu NQR spin-echo technique. We found that $1/T_1$ at the Cu(B)-site is different from that at the Cu(A)-site, and that $1/T_1$ at the Cu(A)-site has a superlinear T-dependence in the T-range of 300 $\sim$ 500 K. Comparisons of the present $1/T_1$ are made with the data for other systems. The difference in $1/T_1$ between two Cu sites is discussed from the viewpoint of local inhomogeneous electronic states.

KEYWORDS: $La_{2-x}Sr_xCuO_4$, high T_c superconductor, NQR, experiment

Most metallic states of the high-T_c Cu oxides arise from doping of holes or electrons into the Mott insulators by cation substitution or oxidation. The chemical doping often produces different local valence states in the metals. A prominent feature is multistructure consisting of clearly separated lines in the Cu nuclear quadrupole resonance (NQR) spectrum observed for $YBa_2Cu_3O_y$ ($6.4 \leq y \leq 6.98$, YBCOy),[1] for Sr-doped $La_{2-x}Sr_xCuO_4$ ($0.15 \leq x \leq 0.50$, LSCO100x%),[2-4] and for oxygen-rich $La_2CuO_{4+\delta}$.[5] These Cu NQR spectra are considered to indicate charge differentiation into several valence states in the CuO_2 plane and are associated with an anomalous metallic property under strong electron correlation.[1] In particular, the metallic LSCO is considered to be a highly correlated system in close proximity to the insulating phase. The nature of the so-called A-, B- and C-lines in the Cu NQR spectrum of metallic LSCO is one of the most intriguing issues.[3,4]

LSCO20% ($T_c \sim$ 30 K) has a nearly T-independent uniform spin susceptibility $\chi(\boldsymbol{q} = \boldsymbol{0})$, similar to that of YBCO6.98 ($T_c \sim$ 92 K).[6-8] The amount of doped holes per CuO_2-sheet, p_{sh}, is estimated to be +0.20 in ionic effective valence, close to the value of $p_{sh} \sim$ +0.24 for YBCO6.98.[9] Therefore, some similar aspects of magnetic behavior in LSCO20% and YBCO6.98 may be expected. In this letter, we report the T-dependence of Cu nuclear spin-lattice relaxation rate $1/T_1$ at the Cu(A)-site (77.3 K $\leq T \leq$ 500 K) and at the Cu(B)-site (77.3 K $\leq T \leq$ 300 K) for LSCO20% ($T_c \sim$ 30 K) above T_c, in order to study the spin dynamics at the A- and B-sites and to compare it with that of the optimum-doped LSCO15% ($T_c \sim$ 38 K) or YBCO6.98. We believe that this Sr-content system is free from the structural-phase instability.

A polycrystalline sample of LSCO20% was prepared by the standard solid-state reaction method, following ref. 3. The sample was confirmed by X-ray diffraction to be a single phase. The value of T_c was 30 K, determined from the change of T-dependence of dc-electrical resistivity for a sintered pellet and from the change in coil inductance of a NQR probe for the powdered sample. The zero-field NQR experiments were carried out using the spin-echo technique with a coherent-type pulsed spectrometer. The Cu nuclear spin-lattice relaxation rate $1/T_1$ was measured using the saturation recovery technique.

The observed Cu NQR spectrum at 31 $\sim$ 42 MHz was the same as those in refs. 3 and 4. The NQR frequency of each line was nearly T-independent in the measured T-range. Then, measurements of $1/^{63}T_1$ for the ^{63}Cu(A) and the ^{63}Cu(B) lines were made at 36.6 MHz and 39.2 MHz, respectively, throughout the entire T-range. The experimental recovery curve followed a theoretical single exponential function for the A-line ($T >$ 100 K) and the B-line ($T >$ 125 K). The relaxation mechanism is confirmed to be magnetic for the Cu(A) site by the isotope-dependence of $^{63}T_1$ and $^{65}T_1$. Unfortunately, the recovery curve at low temperatures is of multi-exponential type. Below 100 K, the signal of ^{63}Cu(A) at 36.6 MHz is overlapped by the ^{65}Cu(B) signal, so that a minor component with a long time constant T_{1L} is considered to appear in the above multi-exponential recovery curve. However, this observation of $T_{1L}[\sim(^{63}\gamma_n/^{65}\gamma_n)^2 \times {}^{63}T_1(\mathrm{B})]$ below 100 K ensures that the relaxation mechanism is also magnetic at the ^{63}Cu(B) site. Below 125 K, the signal of ^{63}Cu(B) at 39.2 MHz is disturbed by leakage of the slightly broadened ^{63}Cu(A) line, so that a minor component with a short time constant $T_{1S}[\sim{}^{63}T_1(\mathrm{A})]$ appears. We will report a short time constant at 36.6 MHz and a long time constant at 39.2 MHz as the $^{63}T_1$ for the A- and the B-sites at low temperatures, respectively.

In Fig. 1, the values of $1/^{63}T_1$ for ^{63}Cu(A) (•) and for ^{63}Cu(B) (○) are plotted against T. One can see a difference in $1/^{63}T_1$ between the two sites, namely, $1/^{63}T_1$(A) is 1.2 $\sim$ 2.1 times larger than $1/^{63}T_1$(B). Within the T-range of 300 $\sim$ 500 K, one can also see a superlinear T-dependence of $1/^{63}T_1$(A) similar to that for the pla-

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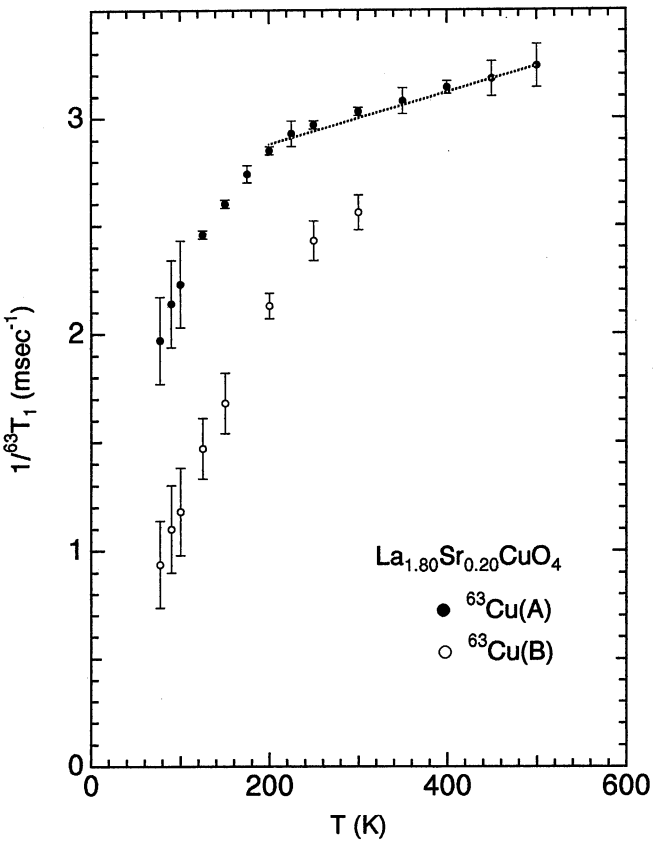


Fig. 1. The ^{63}Cu nuclear spin-lattice relaxation rates $1/^{63}T_1$ at the ^{63}Cu(A) site (•) and at the ^{63}Cu(B) site (○) are plotted against T. The dashed line is a least-squares fit by a function of $a + bT$.

nar Cu site in YBCO6.98.[1] The dashed line is a result of least-squares fitting using a function of $a + bT$ with $a \cong 2.6\,\mathrm{msec}^{-1}$ and $b \cong 1.2\,\mathrm{sec}^{-1}\,\mathrm{K}^{-1}$. Here, we note that the magnitude of $1/^{63}T_1$(A) is larger than those previously reported for the optimum-doped LSCO15%[6,10] and for LSCO20%.[6] The $1/^{63}T_1$ for LSCO20% reported in ref. 6 is smaller than our $1/^{63}T_1$(A) but larger than our $1/^{63}T_1$(B). The reason for this disagreement is not clear. In Fig. 2, the values of $^{63}T_1T$ are plotted against T. $^{63}T_1T$(B) appears to saturate at lower temperatures, in contrast to $^{63}T_1T$(A). The ratio of $1/^{63}T_1$(A) to $1/^{63}T_1$(B) is plotted against T in Fig. 2(b); it decreases with increasing T.

The superlinear behavior of $1/^{63}T_1$(A) at higher temperatures for LSCO20% can be understood by considering the contribution from a nearly T-independent uniform susceptibility $\chi(\boldsymbol{q} = \boldsymbol{0})$ around $150 \sim 400\,\mathrm{K}$[6-8] similar to that of YBCO6.98,[1,8] because the b-term estimated from $\chi(\boldsymbol{q} = \boldsymbol{0})$ was of the same order of magnitude as the experimental one.[11] Thus, the increase of the spin susceptibility $\chi(\boldsymbol{q})$ around $\boldsymbol{q} \sim \boldsymbol{0}$ from LSCO15% to LSCO20%[6,7] can be partially responsible for the increase of $1/^{63}T_1$(A) from LSCO15% to LSCO20%. The other contributions to the increase of $1/^{63}T_1$(A) may be due to increase in the magnitude of the hyperfine coupling constant and/or decrease in the value of the Weiss temperature Θ in the Curie-Weiss-type $1/^{63}T_1T(\mathrm{A}) \sim 1/(T + \Theta)$ below 300 K in Fig. 2(a).[12] That is, the estimated $\Theta_\mathrm{A}(\sim 60\,\mathrm{K})$ for LSCO20% is smaller than the reported $\Theta_\mathrm{A}(\sim 75\,\mathrm{K})$ for LSCO15%. However, we need further experimental data for the planar oxygen NMR to achieve the precise estimation of each contribution from the uniform and the staggered mode. No data are available at present.

Here, we discuss the experimentally observed differences between two Cu sites, i.e., the magnitude of $1/^{63}T_1$(B) is smaller than that of $1/^{63}T_1$(A), while the Cu(B) NQR frequency is higher than that of Cu(A) by about 3 MHz. Let us assume at first a common enhanced spin susceptibility $\chi(\boldsymbol{Q}^*)$ (the incommensurate

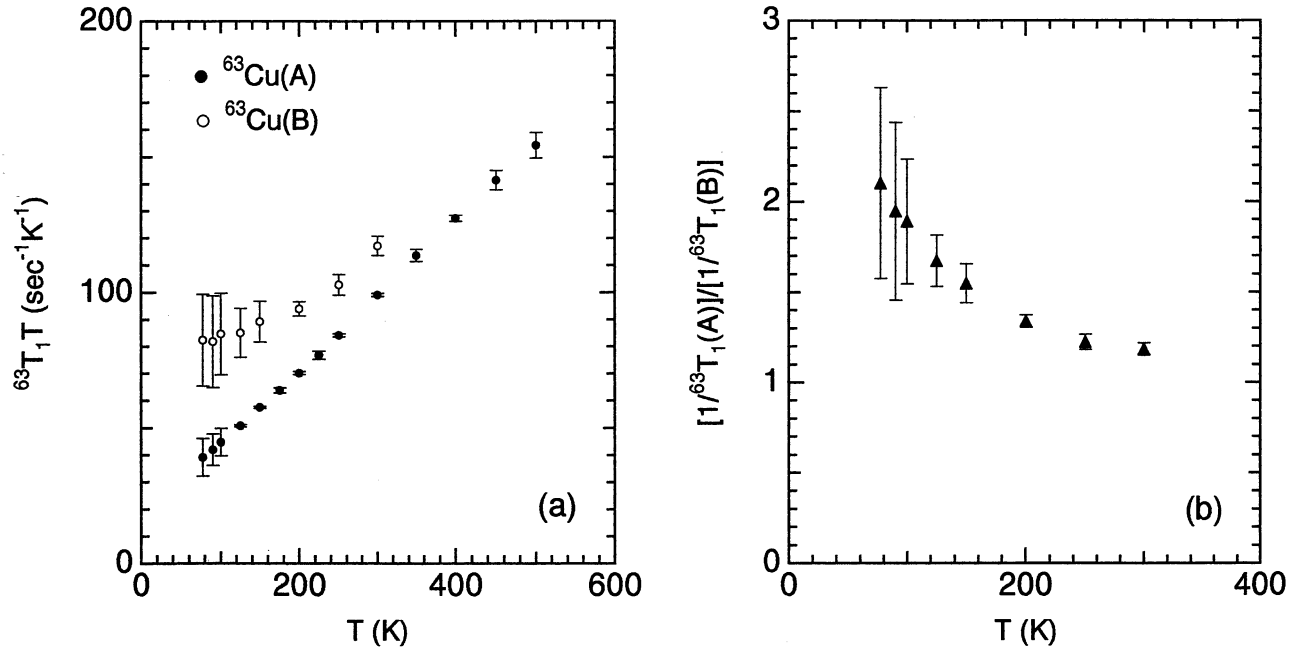


Fig. 2. (a)The relaxation rates $^{63}T_1T$ at the ^{63}Cu(A) site (•) and at the ^{63}Cu(B) site (○) are plotted against T from Fig. 1. (b) The ratio of $[1/^{63}T_1(\mathrm{A})]/[1/^{63}T_1(\mathrm{B})]$ is plotted against T. The ratio can be seen to decrease with increasing T.

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vector $\boldsymbol{Q}^*$) shared between two sites. When the doped Sr introduces a local structural distortion such that the apical oxygen is relocated away from its regular position and then the Cu(B)-site is located close to the unoccupied apical site,[3,4] we can consider that two separated Cu NQR lines arise from a difference in the charge transfer process through the apical oxygen.[13] A change in the charge transfer process may yield a difference in the $\boldsymbol{q}$-dependence of the hyperfine coupling constant $A_\alpha(\boldsymbol{q}) = A_\alpha + 2B[\cos(q_x) + \cos(q_y)]$ ($\alpha = ab$ and cc).[14] Here, A_α and B are the anisotropic on-site hyperfine coupling constant along the α-axis and the isotropic transferred one, respectively.[14] We follow the calculations in ref. 15 to analyze the T-dependent ratio of the ^{63}Cu nuclear spin relaxation rates, $r \equiv \{1/^{63}T_1(\mathrm{A})\}/\{1/^{63}T_1(\mathrm{B})\}$. As in ref. 15, we can expand the ratio r with respect to the antiferromagnetic correlation length $\xi/a(\gg 2)$ as

$$r \cong \frac{(A_{ab}-4B)^2_{A-site}}{(A_{ab}-4B)^2_{B-site}}\left[1+20\left\{\frac{B^2_{B-site}}{(A_{ab}-4B)^2_{B-site}} - \frac{B^2_{A-site}}{(A_{ab}-4B)^2_{A-site}}\right\}L_0(\xi/a)\right]. \quad (1)$$

Here, $L_0(\xi/a)[= (a/\xi)^2\ln(\xi/a)]$ is an increasing function of T,[15] because $(\xi/a)^2[\propto \chi(\boldsymbol{Q}^*)]$ for LSCO20% can be expected to decrease with increasing T in order to reproduce the T-dependence of T_1T in Fig. 2(a). The notation $(\cdots)_i$ (i = A- and B-site) represents each physical quantity at the Cu(A)- and the Cu(B)-site. Since $L_0(\xi/a) \ll 1$ for $\xi/a \gg 2$ in the low-temperature limit ($T \to 0$), we obtain the asymptotic form of eq. (1) as $r \to (A_{ab}-4B)^2_{A-site}/(A_{ab}-4B)^2_{B-site}$ ($T \to 0$). Then, from the experimental ratio $r > 1.0$ ($T \to 0$) in Fig. 2(b), we require $|A_{ab}-4B|_{B-site} < |A_{ab}-4B|_{A-site}$. On the other hand, to explain the decrease in the ratio r with increasing T in Fig. 2(b), we require $\{\cdots\} < 0$ in eq. (1), i.e., $|A_{ab}-4B|_{B-site} > (B_{B-site}/B_{A-site})|A_{ab}-4B|_{A-site}$. To obtain reasonable values of A_{ab} and B consistent with these results, we have a final requirement of $B_{B-site} < B_{A-site}$. That is, the Cu cation-cation electron transfer via the apical oxygen is suppressed at the Cu(B) site. If we associate this reduced B_{B-site} with an increase of the transfer gap Δ, defined in ref. 16, between Cu 3d- and O 2p-states,[17] the Cu(B) NQR frequency higher than that of Cu(A) can be explained by the cluster calculations in ref. 16.

However, we should also discuss the possibility that the local spin susceptibility $\chi_B(\boldsymbol{Q}^*)$ at the Cu(B)-site is not shared with the Cu(A)-site under the condition of $[A_\alpha(\boldsymbol{q})]_{B-site} \cong [A_\alpha(\boldsymbol{q})]_{A-site}$. In Fig. 2(a), we can approximate a Curie-Weiss-type $\chi_B(\boldsymbol{Q}^*) \sim 1/(T+\Theta_B)$ with $\Theta_B \sim 160\,\mathrm{K}$ at $200 \sim 300\,\mathrm{K}$ and $\chi_A(\boldsymbol{Q}^*) \sim 1/(T+\Theta_A)$ with $\Theta_A \sim 60\,\mathrm{K}$ at $100 \sim 300\,\mathrm{K}$.[12,14] If we consider that the large value of Θ_B results from a local electron correlation weakened by highly concentrated ("rich") holes,[12] the Cu(B)- and the Cu(A)-sites can be assigned to the hole-rich and the hole-poor states, respectively.[18,19] The ratio of $^{63}T_1(\mathrm{B})/^{63}T_1(\mathrm{A})$ expressed by $1+(\Theta_B-\Theta_A)/(T+\Theta_A)$ with $\Theta_B > \Theta_A$ easily reproduces the T-dependence in Fig. 2(b). Here, we can adopt the following two viewpoints for the role of the doped Sr in the LSCO system; (1) a pinning center for an inhomogeneous charge distribution with hole-rich and hole-poor states,[18,19] or (2) a center of modulation of the charge distribution. The experimentally observed relationship between the amount of doped Sr and the relative intensities of respective NQR lines may be a key information to clarify these points.[3,4] Further experiments are being considered to clarify the microscopic nature of the inhomogeneous electronic states and their systematic variation.

In conclusion, we found for LSCO20% a difference in the magnitude and the T-dependence of $1/^{63}T_1$ between the Cu(A)- and the Cu(B)-sites, and a superlinear T-dependence of $1/^{63}T_1$ at the Cu(A)-site in the T-range of $300 \sim 500\,\mathrm{K}$, similar to that of $1/^{63}T_1$ at the planar Cu site in YBCO6.98. We observed an increase of $1/^{63}T_1(\mathrm{A})$ from LSCO15% to LSCO20%. There are three possible explanations for this increase: an increase of $\chi(\boldsymbol{q} \sim \boldsymbol{0})$, an increase in the magnitude of the hyperfine coupling constant, and/or an increase of the Weiss temperature for a Curie-Weiss-type $\chi(\boldsymbol{Q}^*)$.

We thank Professor K. Yoshimura and Professor H. Yasuoka for stimulating discussions. This work has been partially supported by a Grant-in-Aid for Scientific Research from the Ministry of Education, Science and Culture. One of the authors (Y.I.) has been supported by a Research Fellowship of the Japan Society for the Promotion of Science for Young Scientists.

1) H. Yasuoka, T. Imai and T. Shimizu: *Strong Correlation and Superconductivity*, ed. H. Fukuyama, S. Maekawa and A. P. Malozemoff (Springer, Berlin, 1989) p. 254.
2) K. Kumagai and Y. Nakamura: Physica C **157** (1989) 30.
3) K. Yoshimura, T. Imai, T. Shimuzu, Y. Ueda, K. Kosuge and H. Yasuoka: J. Phys. Soc. Jpn. **58** (1989) 3057.
4) K. Yoshimura, T. Uemura, M. Kato, K. Kosuge, T. Imai and H. Yasuoka: Hyperfine Interactions **79** (1993) 867.
5) P. C. Hammel, A. R. Reyes, S-W. Cheong and Z. Fisk: Phys. Rev. Lett. **71** (1993) 440.
6) S. Ohsugi, Y. Kitaoka, K. Ishida, G.-q. Zheng and K. Asayama: J. Phys. Soc. Jpn. **63** (1994) 700.
7) L. F. Schneemeyer, J. V. Waszczak, E. A. Rietman and R. J. Cava: Phys. Rev. B **35** (1987) 8421.
8) D. C. Johnston, S. K. Sinha, A. J. Jacobson and J. M. Newsam: Physica C **153–155** (1988) 572; R. Yoshizaki, N. Ishikawa, H. Sawada, E. Kita and A. Tasaki: Physica C **166** (1990) 417; T. Nakano, M. Oda, C. Manabe, N. Momono, Y. Miura and M. Ido: Phys. Rev. B **49** (1994) 16000.
9) Y. Tokura, J. B. Torrance, T. C. Huang and A. I. Narral: Phys. Rev. B **38** (1988) 7156; J. B. Torrance, A. Bezinge, A. I. Nazzal, T. C. Huang, S. S. P. Parkin, D. T. Keane, S. J. LaPlaca, P. M. Horn and G. A. Held: Phys. Rev. B **40** (1989) 8872.
10) T. Imai, C. P. Slichter, K. Yoshimura and K. Kosuge: Phys. Rev. Lett. **70** (1993) 1002; M. Matsumura, Y. Itoh, H. Yasuoka, Y. Ueda and H. Yamagata: unpublished works.
11) For LSCOx%($18 \le x \le 20$), we roughly estimated the b-term to be $1.9 \sim 2.5\,\mathrm{sec}^{-1}\mathrm{K}^{-1}$ from the reported $\chi(\boldsymbol{q}=\boldsymbol{0})$,[6-8] where we used the T-independent spin part $\chi^0[= 0.6 \sim 0.8 \times \chi(\boldsymbol{0})]$ of the reported $\chi(\boldsymbol{0})$.[6-8] Here, we assumed a nearly $\boldsymbol{q}$-independent background spin susceptibility $\chi_{BG}(\boldsymbol{q})[\cong \chi(\boldsymbol{0})]$ and appropriate hyperfine coupling constants, similarly to [Y. Itoh and H. Yasuoka: J. Phys. Soc. Jpn. **63** (1994) 2518].
12) Y. Itoh: preprint submitted to *International Symposium, Frontiers of High T_c Superconductivity*, 1995, Morioka, Iwate, Japan (to appear in Physica C); Y. Itoh: J. Phys. Soc. Jpn.

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63 (1994) 3522 [Erratum: **64** (1995) 684].
13) H. Matsukawa and H. Fukuyama: J. Phys. Soc. Jpn. **59** (1990) 1723.
14) F. Mila and T. M. Rice: Physica C **157** (1989) 561; B. S. Shastry: Phys. Rev. Lett. **63** (1989) 1288; A. J. Millis, H. Monien and D. Pines: Phys. Rev. B **42** (1990) 167.
15) Y. Itoh, A. Hayashi and Y. Ueda: J. Phys. Soc. Jpn. **64** (1995) 3072.
16) Y. Ohta, W. Koshiba and S. Maekawa: J. Phys. Soc. Jpn. **61** (1992) 2198.
17) E. Simánek and Z. Sroubek: *Electron Paramagnetic Resonance* ed. S. Geschwind (Plenum Press, New York, 1972) Chap. 8, p. 535.
18) R. L. Martin : Phys. Rev. Lett. **75** (1995) 744.
19) V. J. Emery, S. A. Kivelson and H-Q. Lin: Phys. Rev. Lett. **64** (1990) 6523; V. J. Emery and S. A. Kivelson: Physica C **209** (1993) 597; J. M. Tranquada, B. J. Sternlieb, J. D. Axe, Y. Nakamura and S. Uchida: Nature **375** (1995) 561.